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SUPERSEDING
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18 May 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7256

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, tunable frequency range - 9,000 to 9,160 MHz, integral magnet, 40-kW rated peak-power output

ABSOLUTE RATINGS:

Parameter:	Ef	tk	tpc	Du	ib	epy	Pi	VSWR	Tuner torque	Altitude
Unit:	V	sec	µsec	---	a	kV	W	----	in-oz	feet
Maximum:	7.0	---	0.6	.0011	15.5	16	240	1.5	40	20,000
Minimum:	---	120	---	----	12.5	--	---	---	--	-----

PHYSICAL CHARACTERISTICS: - (See figure 1)

Mounting:	Mounting flange	Input coupling:	See figure 1
Mounting position:	Any	Weight:	5 lbs (approximately)
Output coupling:	See note 3	Cooling:	Air

TEST CONDITIONS:

Parameter:	Ef	tk	Du	tpc	trv	Ib	VSWR	Frequency
Unit:	V	sec	--	µsec	µsec	mAdc	----	MHz
Tolerance:	---	(max)	--	±0.05	(min)	--	----	---
Test Condition 1:	6.3	120	--	---	---	--	----	---
Test Condition 2:	0	---	.0010	0.5	0.08	14	1.1:1	F_1
Test Condition 3:	0	---	.0010	0.5	0.08	14	1.1:1	F_1, F_2, F_3
Test Condition 4:	0	---	.0010	0.5	0.08	14	1.1:1	F_3

TEST FREQUENCIES
F_1 = 9,000 MHz
F_2 = 9,080 MHz
F_3 = 9,160 MHz

GENERAL:

Qualification: - Required

Ⓑ Denotes changes

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1		See Note 1				
1301	Heater current	1		If	0.90	1.10	A
4003	Pressurizing		40 lb_f/in^2 absolute (input and output)	---	----	----	---
4306	Pulse voltage	2		epy	13	15	kV
4250	Power output	3		Po	40	----	W
4308	RF bandwidth	3		BW	----	2.0/tpc	MHz
4308	Minor lobes	2		ratio	6	----	dB
4315	Starting stability	2	See Note 4	MP	----	1.0	%
	Quality conformance inspection, part 2		See Note 2				
1031	Low frequency vibration		No voltages	---	----	----	---
(B) 4223	Mechanical tuning range	3	High frequency limit	F	9160	----	MHz
			Low frequency limit	F	----	9,000	MHz
4223	Tuner operation torque		TA = 30°C (max)	Torque	----	10	in-oz
4308	RF bandwidth	3	VSWR = 1.5:1 Ib = 12.5 & 15.5 mAdc See Note 5	BW	----	2.2/tpc	MHz
----	Reverse tuning	3	See Note 6	ΔF	----	200	kHz
4310	Pulling factor	2		---	----	----	---
4315	Stability	3	VSWR = 1.5:1 Ib = 15.5 mAdc See Note 7	MP	----	0.5	%
	Quality conformance inspection, part 3						
(B) 4551	Cycled life test	3	VSWR = 1.5:1 (min) See Note 11	N	223	----	cycles
(B) 4551	Life-test end points:						
4250	Power output	3		Po	32	----	W
4306	Pulse voltage	2		epy	12	16	kV
4315	Stability	3	Ib = 15.5 mAdc See Note 7	MP	---	1.0	%
4308	RF bandwidth	2	Ib = 12.5 & 15.5 mAdc	BW	----	2.5/tpc	MHz
4308	Minor lobes	3		ratio	6	----	dB
1031	High frequency vibration		No voltages See note 12	---	----	----	---
1041	Shock		See Notes 8 and 12	---	----	----	---

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued						
1143	Air cooling	2	TA = 50°C (max) Pi - Po = 150 W See Notes 9, 12 and y	ΔT	---	TA+50	°C
4223	Tuner operating torque		TA = -55 & +125°C See Note 12	Torque	---	32	in-oz
----	Anode - cathode cap capacitance			C	4.0	8.0	pF
4223	Mechanical fatigue		See Notes 10 and 12	N	2,500	---	cycles
4027	Temperature coefficient	4	T(anode) = 70 to 100°C See Notes 12 and y	ΔF/ΔT	---	0.25	MHz/°C
1002	Barometric pressure reduced		Pressure = 349 mmHg F = 9,160 MHz See Note 12	---	---	---	---
1047	Low temperature operation	4	Stability See Note 7 and 12	MP	---	2	%

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under Quality Conformance Inspection, Part 1, shall be 1.0% with inspection level of II.

2. Unless otherwise specified, the AQL for all tests listed under Quality Conformance Inspection, Part 2, shall be 6.5 with inspection level of S3.

3. The magnetron may be coupled directly to a RG-51/U waveguide in accordance with MIL-W-85, with a UG-52B/U choke flange in accordance with MIL-F-3922/25. However, for a minimum VSWR of this coupling, it is recommended that a specially designed choke flange be used which mates with the angular RG-51/U output of the magnetron to provide a straight section of RG-51/U waveguide through the coupling network.

4. Immediately after the holding period, and after a 120-second preheat, the pulse voltage as specified in Quality Conformance Inspection, Part 1, shall be instantaneously applied and the magnetron missing pulse count shall not exceed 1 percent over the first minute of operation.

5. The RF bandwidth shall be within the limits specified when a VSWR of 1.5 is introduced in the load at a distance no greater than 1/2 meter from the magnetron coupling flange, the phase being adjusted for the worst spectrum.

6. When the direction of tuner drive is reversed at the frequency specified, the magnetron shall not continue to tune in the original direction by more than the specified frequency increment before reversing. Test to be performed at 9,000 MHz, and 9,080 MHz, and 9,160 MHz.

7. Stability shall be measured with the VSWR adjusted to the phase which produces maximum instability. The missing pulses shall be counted during any 3 minute period of a test interval not to exceed 6 minutes.

8. The tube shall be subjected to five shocks of peak amplitude duration in accordance with MIL-STD-202, Method 213, Test Condition G. Following the shock test, the tube shall show no mechanical damage, and shall pass the power output and pulse voltage requirements of Quality Conformance Inspection, Part 1.

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NOTES: -Continued

9. An airflow, at standard atmospheric pressure, of 25 ft^3/min shall be directed at the cooling fins from an orifice of 1.1719 inches (2.977 mm) by 1.3594 inches (34.53 mm) located 0.250 inches (6.35 mm) from the cooling fins.

10. The tuning shaft shall be continuously driven at a speed of approximately 650 r/min. The tube shall meet end-of-life requirements at completion of this test.

11. Life test shall be conducted in accordance with the following cycle:

Test condition	Duration
1	2 minutes
3	4 hours
off	28 minutes

(R) 12. This test need not be performed if last performed within 3 years and no related change has occurred in design, materials used or production methods used in the manufacture of tube type 7256.

(B) 13. The use of reclaimed materials shall be encouraged to the maximum extent possible.

Custodians:
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Navy - EC
Air Force - 85

Preparing activity:
Air Force - 85

Review activities:
Navy - SH
Air Force - 99
DLA - ES

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User activities:
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Air Force - 11

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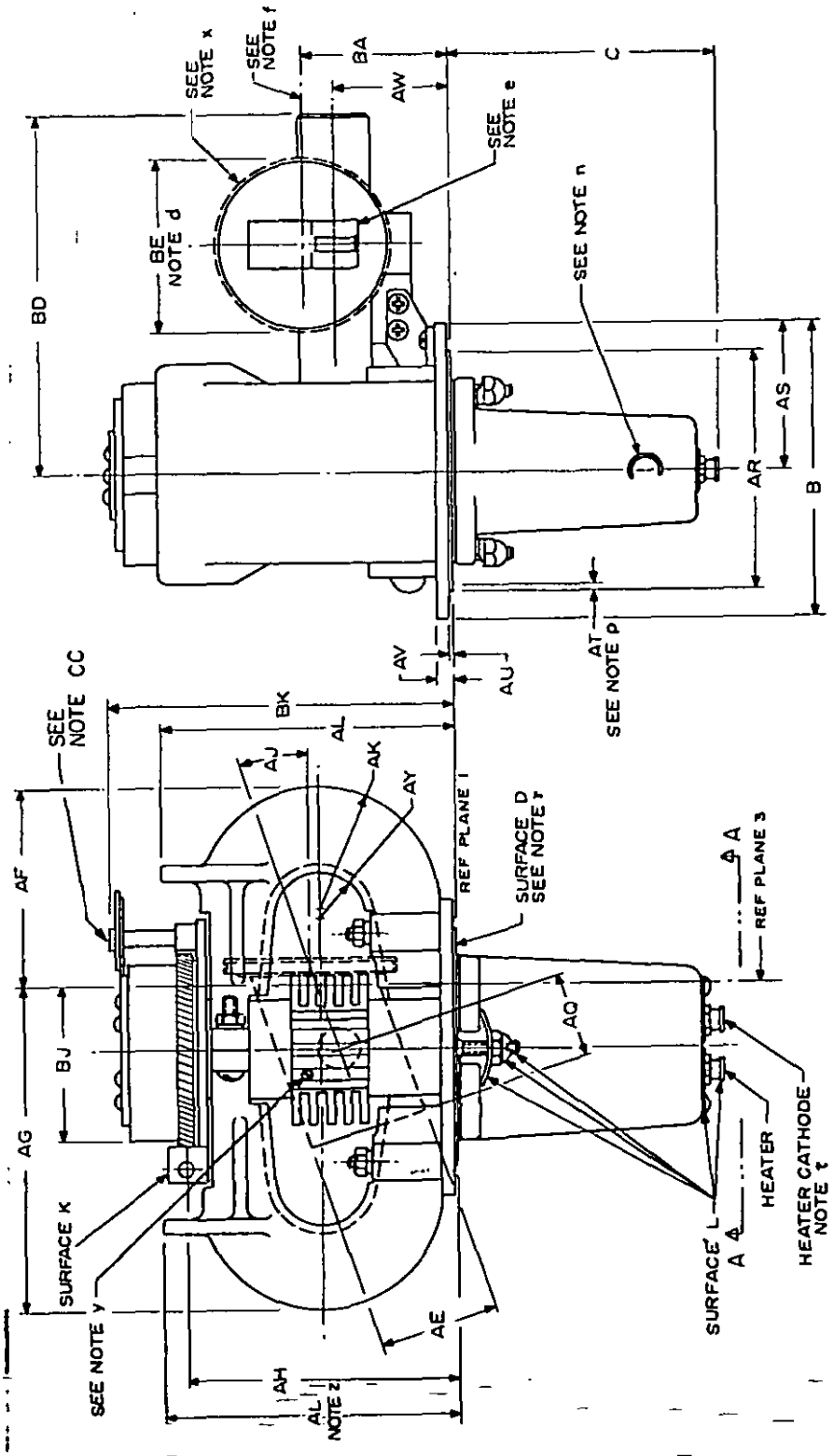


FIGURE 1. Outline drawing of electron tube type 7256

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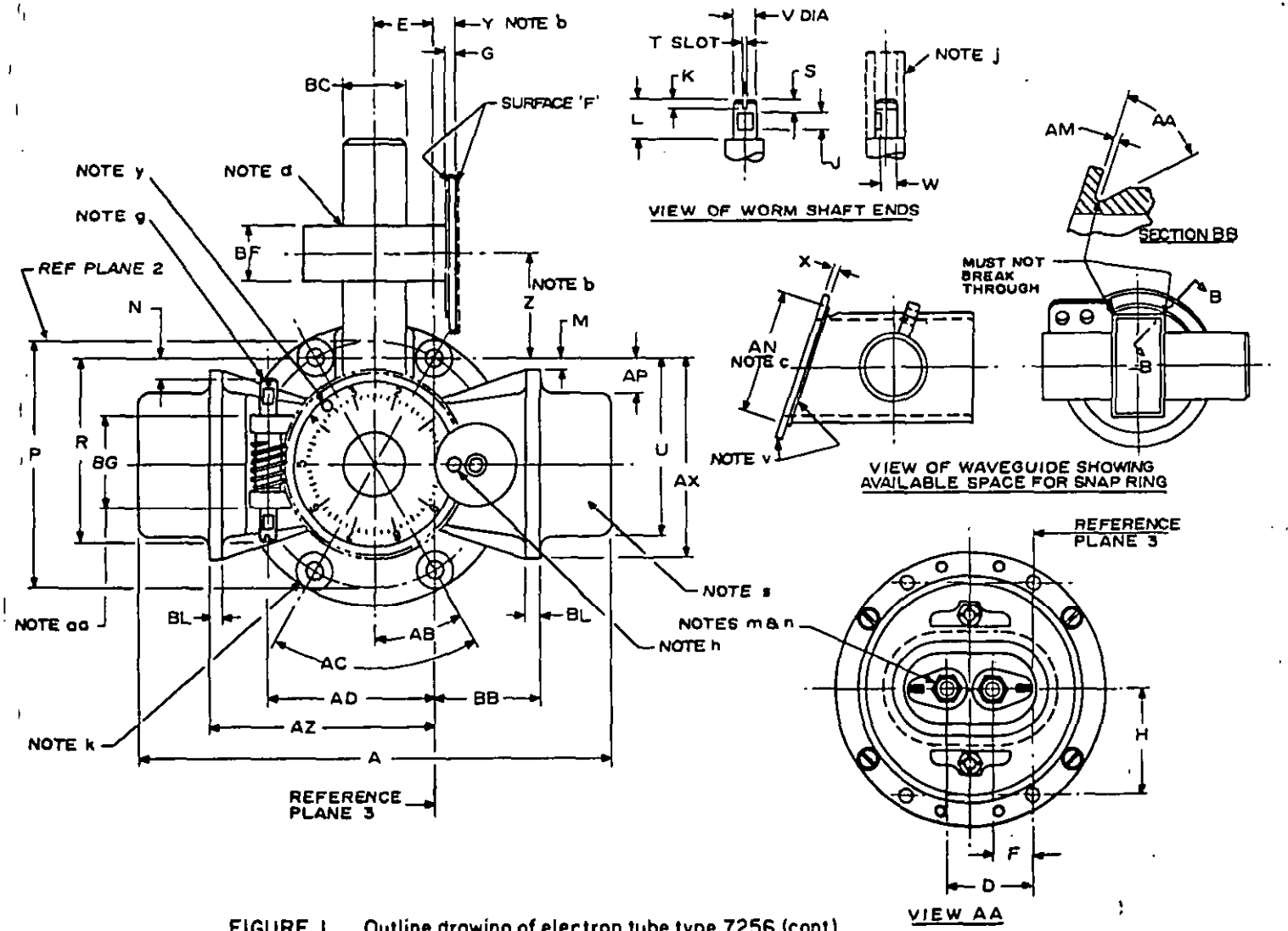


FIGURE 1 Outline drawing of electron tube type 7256 (cont)

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DIMENSIONS

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECT., PT. 1				
D	.969	Nom	24.61	
E	.719	Nom	18.26	
F	.469	Nom	11.91	
H	1.245	Nom	31.62	
K	.115	.135	2.92	3.43
M	.042	-	1.07	-
N	.230	.290	5.84	7.37
R	2.105	2.165	53.47	54.99
Y	.261	.301	6.63	7.65
Z	1.173	1.213	29.79	30.81
AD	1.954	1.984	49.63	50.39
AH	2.834	2.874	71.98	72.00
AM	.021	.031	.53	.79
AN	1.438	1.446	36.53	36.73
AX		2.453		62.31
BA	1.542	1.582	39.17	40.18
BE	1.743	1.757	44.27	44.63
BG	.990	1.010	25.15	25.65
QUALITY CONFORMANCE INSPECT., PT. 2				

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A		5.937		150.80
B	3.219	3.281	81.76	83.34
C	2.923	3.047	73.24	77.39
J	.187	.203	4.75	5.16
L	.428	.448	10.67	11.38
P	2.869	2.881	72.87	73.18
S	.140	.156	3.56	3.96
T	.040	.045	1.02	1.14
U		2.187		55.55
V	.1865	.1895	4.74	4.81
AC	59°48'	60°12'		
AF		2.250		57.15
AG		3.687		93.65
AL		3.281		83.34
AP	.308		7.82	
AR	2.620	2.630	66.55	66.80
AS	1.625	Nom	41.28	
AT	.057	.067	1.45	1.70
AZ		2.781		70.64
BB		1.343		34.11
BD		3.906		99.21

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
BF	.625	Nom	15.88	
BJ		1.700		43.18
BK		4.200		106.70
BL	.1875	Nom	4.76	
QUALITY CONFORM. INSP. PT. 3, NOTE 12				
G	.125	Nom	3.18	
Q	1.438	Nom	36.53	
W	.151	.161	3.84	4.09
X	.080	.090	2.03	2.29
AA	45°			
AB	30°			
AE	1.250	Nom	31.75	
AJ	18°42'			
AK	1.296	Nom	32.92	
AQ	.930	Nom	23.62	
AU	.023	.039	.58	.99
AV	.156		3.96	
AW	1.224	Nom	31.09	
AY	.484	Nom	12.29	
BC	.750	Nom	19.05	

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NOTES:

a. All solder joints on waveguide assembly shall be soldered to provide a hermetic seal at surface (F).
b. The ±.020 (.51 mm) tolerance of .281 (7.14 mm) dimension includes angular as well as lateral deviations of the surface. These dimensions define relation of surface (F) to respective holes.
c. This diameter shall be concentric with the diameter of the flange within .005 (.13 mm).
d. This diameter shall be concentric with opening in waveguide within .010 (.25 mm).
e. The opening in the waveguide shall be enclosed by a dust cover when the tube is not in use.
f. Center line of waveguide opening.
g. To increase frequency, drive this end of worm shaft in counter clockwise direction. Complete frequency range covered in approximately 20 turns. Tuning mechanism shall operate over the entire mechanical range when subjected to a torque of 10 inch-ounces applied at the worm shaft. In equipment no less than 10 inch-ounces nor more than 2,500 inch-pounds shall be applied at drive shaft. The gear teeth and worm threads must be free from corrosion, paint, and other imperfections.
h. Number appearing here indicates number of complete revolutions of gear. With the geneva and gear set near 3 and zero respectively the frequency of the tube is approximately 9000 MHz.
j. It shall be possible for a sleeve .195 (4.95 mm) I.D. x .406 (10.31 mm) O.D. x 1.000 (25.40 mm) long to pass over ends of shaft to face of worm shaft housing.
k. Four .193 ±.003 (4.90 ±.08 mm) diameter holes. An area of .219 (5.56 mm) radius located from the center of each of the four holes to be free from finish.
m. The jack holes shall be within a radius of .023 (.58 mm) of the specified location but shall be spaced .500 ±.010 (12.70 ±.25 mm) with respect to each other, the center lines of the jack holes shall be perpendicular to mounting plate within 3°.
n. 'C' indicates adjacent jack is the common heater-cathode connection.
p. With .062 (1.57 mm) seat resting on a plane surface the flatness of the seat shall be such that a .010 (.25 mm) thickness gage .125 (3.18 mm) wide shall not pass under.
q. All solder joints on base plate shall be soldered to provide an hermetic seal.
r. Any portion of the assembly extending below this surface shall be within a 1.109 (28.17 mm) radius on the true center of the plate.
s. All metal surfaces covered by finish except surfaces (D) and (F) drive and stop mechanism (K) surface around holes as specified in note 10, parts associated with insulator (L).
t. Locking type jack - see latest issue of MIL-E-1.
u. Hex head banana pin jack .594 (15.09 mm) long, hole .169 ±.005 (4.29 ±.13 mm).
v. Finish may be omitted from the edges and back of waveguide.
w. Warning - maintain minimum 2 inches (50.8 mm) clearance between this magnet and magnetic materials (magnets, steel tools, plates, etc.).
x. Waveguide cover required for shipping purposes only.
y. Anode temperature to be measured at this point.
z. Includes top of worm gear housing.
aa. This dimension applies from end of ears of worm shaft housing to a line .190 (4.83 mm) minimum from centerline of worm shaft toward reference plane 3.
bb. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
cc. Geneva cover retained by 6-32/.250 pan head screw and shakeproof washer.

FIGURE 1. Outline drawing of electron tube type 7256 (Cont).

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